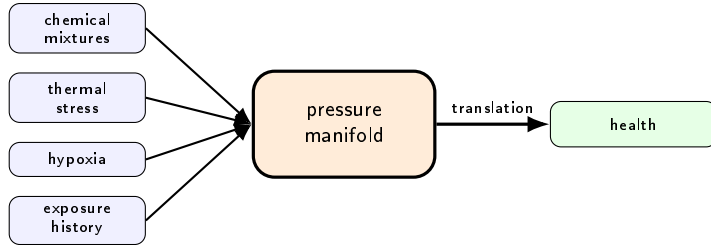


Adaptive margin

Stijn H. Peeters

June 9, 2026

How do we construct a pressure manifold: a many-input device that funnels qualitatively different burdens *into* an account of health?



without resorting to *regulatory* limits as the *stand-in* for “health”.

Possible solution: framing health in energetic terms.

But what is health?

Canguilhem: health as normative capacity



Health is normativity

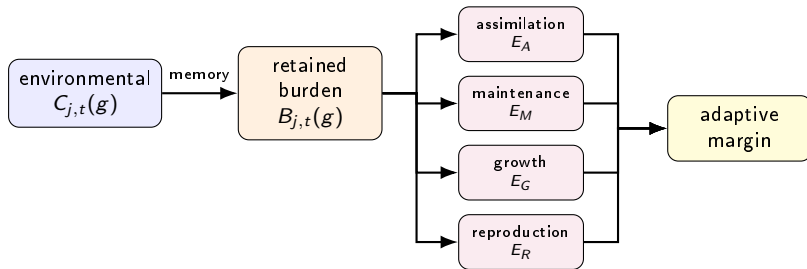
The ability of an organism to change it's
state of being

Main question

How do we go from concentrations to loss
of norm-generating capacity?

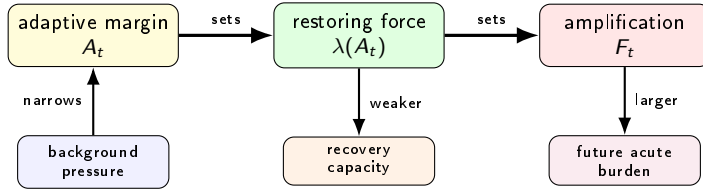
Dynamic Energy Budget Theory

We can map stresses into DEB-like models to estimate remaining adaptive margin, the energetic latitude of the organisms to respond to the environment.



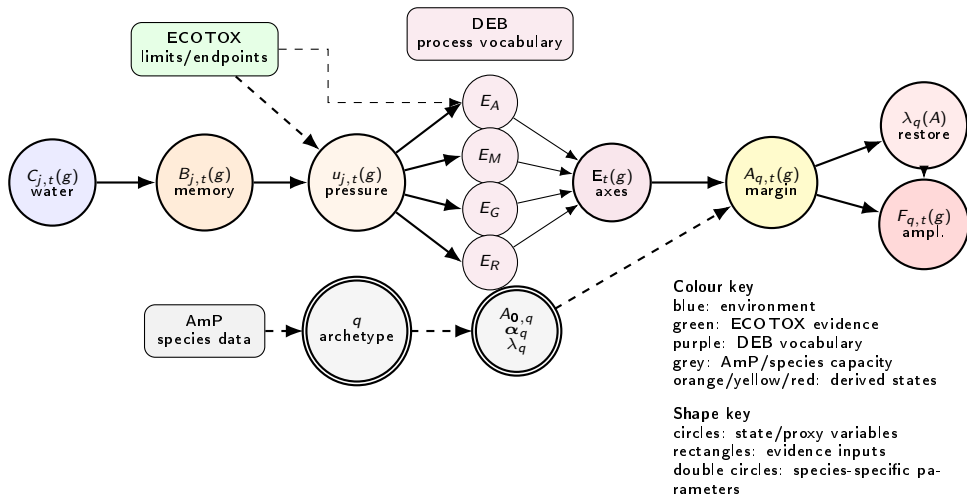
We thus define health as the capacity of an organism to respond to the environment.

From adaptive margin to amplification

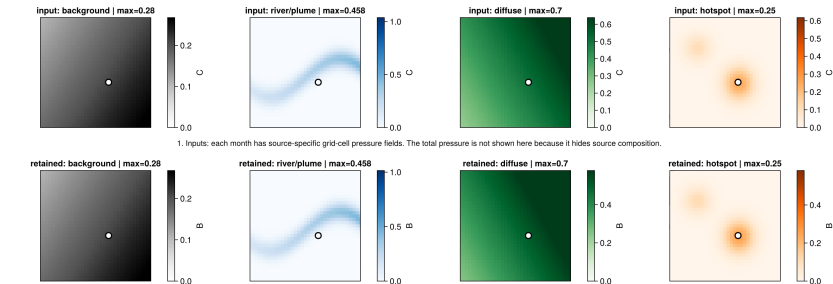


Lower adaptive margin means weaker recovery; weaker recovery means a larger burden from the same acute event.

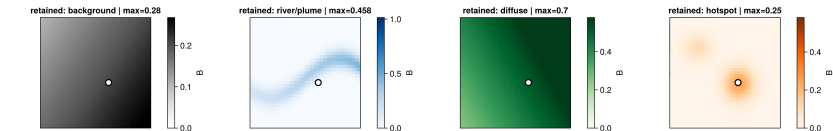
Environment, evidence, and adaptive margin



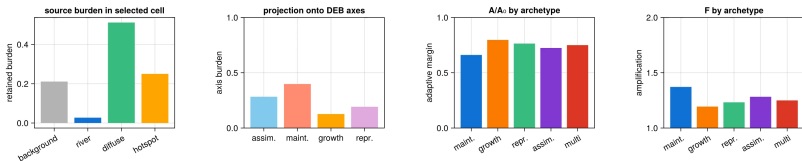
Animated source-resolved story view — month 1, cell (22, 15)



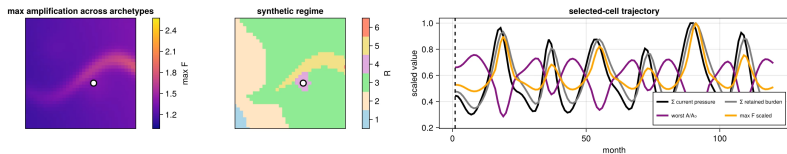
1. Inputs: each month has source-specific grid-cell pressure fields. The total pressure is not shown here because it hides source composition.



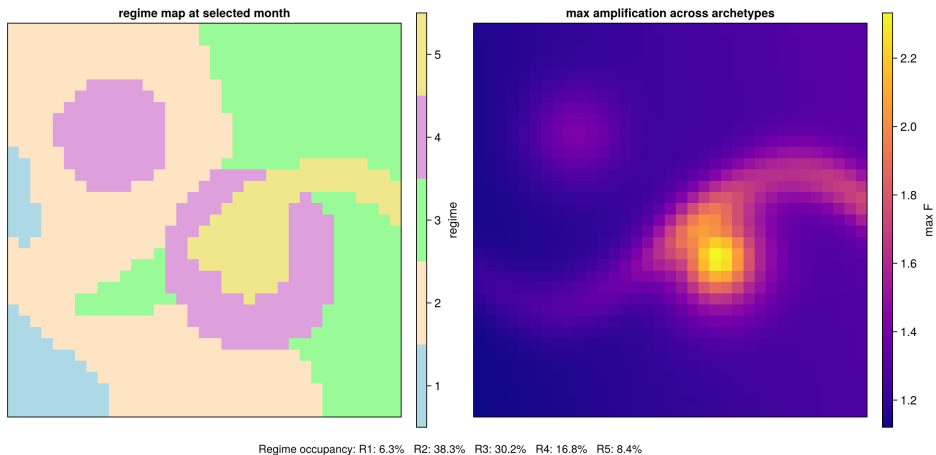
2. Memory: current pressure is carried forward as retained burden. Pulses can persist after the input field has weakened.

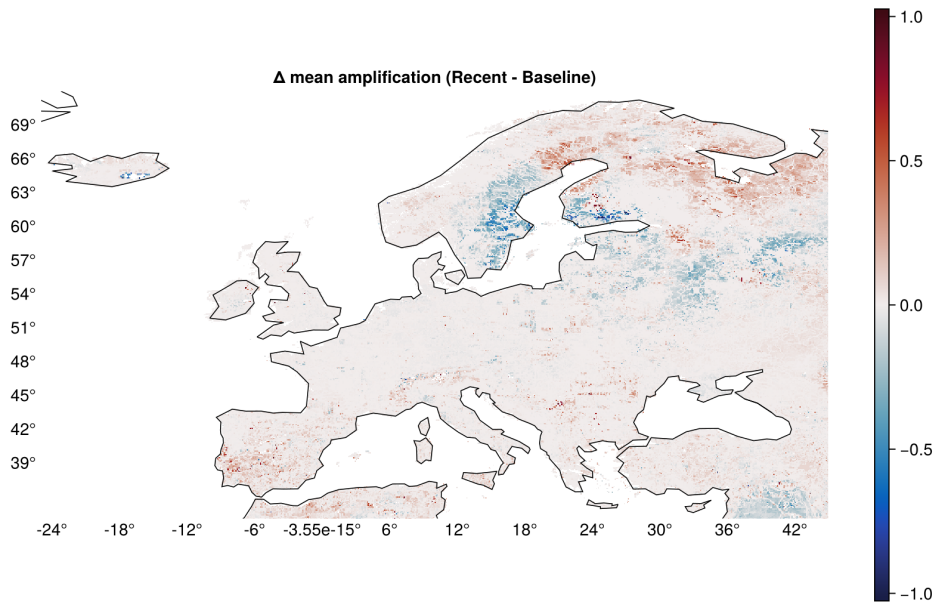


3. The selected-cell retained burden has a source composition. That composition is projected onto energetic axes, then weighted by each archetype.



Time-varying synthetic regimes — month 26, $k = 5$





Closing

*Mapping water quality into Dynamic Energy Budget theory
allows us to:*

- ▶ Combine qualitatively different stresses into a common currency, the adaptive margin.
- ▶ Make use of a *positive* definition of health which applicable to any species.
- ▶ Mathematically useful, regardless of philosophical stance.
- ▶ ... with the caveat that it's new and not yet calibrated.